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Real-Time Speech Emotion Recognition Using Deep Learning for Emotion Based Music Recommendation

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***Abstract*— Affective state is a key factor affecting the music preference and the user experience. Traditional music recommendation systems do not have the capability of emotional sense, and the recommendations may then be contradictory to the user who wants to listen to music in a certain mood. This paper introduces a real-time Speech Emotion Recognition (SER) system, which employs a hybrid model of a CNN and a BiLSTM to recognize the user's emotions from speech input. The model is trained on public available databases--CREMA-D, and RAVDESS--and Mel-Frequency Cepstral Coefficients (MFCC) features with 50 dimensions are used. The system is tailored for mobile implementation with TensorFlow Lite and packaged in a native Android app. Emotion recognition outcome plays emotion-based music according to a pre-stored play-list. The performance of the system is tested under different real-world scenarios ranging from noisy to silent conditions. User testing has shown that users can decern between positive, neutral and negative emotions. In addition, the mobile app possessed good usability and learned user satisfaction, which further implied that it had the possibility to be used to facilitate emotion efficiently using personalized music experiences.**

Keywords*— *Speech Emotion Recognition (SER), Deep Learning, CNN-BiLSTM, Mobile Application, Emotion-Based Music Recommendation, TensorFlow Lite, MFCC, Real-Time Processing, Android App

I. INTRODUCTION

The links between music and emotion have long fascinated psychologists and technologists. Music can strongly modulate emotions in a variety of ways, and can play an important role in human mood and well-being [1]. However, most traditional music recommendation systems build on fine-grained user data or historical listening records, and overlook a user's current emotional status [2]. Such a restriction may cause a poor user experience, the recommended songs are not interactive with the user's sentiment/current mood at all.

In the recent time, solutions for SER have created new avenues for designing intelligent, emotion-sensitive systems. SER is the term for the identification and classification of human emotions from speech signals and gives a more profound, real-time insight into a user's affective condition [3]. When driven by deep learning paradigms such as Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs), SER systems have been reported as being capable of driving up the classification accuracy and the overall generalisation ability across diverse acoustic environments [4].

Deep learning models, especially CNNs, are outstanding in extracting features directly from the raw inputs, utilizing the multiple layers of filters to capture local patterns in spectrograms and MFCC features. When complemented with Bi-directional LSTM (BiLSTM) networks that can efficiently learn temporal dependencies in audio data, these hybrid models are good candidates for SER applications [5].

However, building in SER into mobile applications introduces a particular set of problems. These reasons include low computation resources, background noise and real-time inference requirement. Furthermore, limited number of current applications are using SER for dynamically suggesting music that suits the user's mood. Existing entertainment system or virtual assistants do not have affective intelligence built in them, while studies show that user experience and satisfaction can significantly be increased if this emotional adaptability is integrated into these systems [6].

The paper presents a real-time on-device speech emotion recognition system using CNN-BiLSTM architecture with TensorFlow Lite optimization. The system is incorporated into an Android application that records a user's speech, classifies the emotional state and suggest music according to the user's mood. Classification targets three emotional states (positive, neutral, negative) and allows for real-time personalization of playlists.

The rest of this paper is structured as follows. The methods for data acquisition, preprocessing, model training, and the mobile deployment are detailed in Section II. The mobile system architecture and the emotion-matching music recommendation logic is described in Section III. Section IV presents the experimental results and user feedback. Finally, Section V concludes the paper and discusses future work in emotion granularity, multilingualism and ethics.

II. METHODOLOGY

This part presents the details of the development of the Speech Emotion Recognition (SER) real time system to be used in emotion-based music recommendation. The diagnostic-assessment process consists of data source acquisition, pre-processing data, model and model training, app development, and final system evaluation. Every steps adheres to actual

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mobile constraints, guaranteeing users have best experience and system runs smoothly

A. Dataset Selection and Preprocessing

Three existing, publicly accessible emotional speech datasets (TESS, RAVDESS and CREMA-D) were used to train and test the model. These databases provide various emotional utterances from several speakers between young to elderly age and with various backgrounds which enhance the generalization capability of the model.

The dataset was generalized to three primary emotional categories in speech: Positives, Neutrals, Negatives in order to facilitate easy real time mobile inference. Emotion mapping was done using Happy, Calmed, and Surprised as positive Neutral as neutral and Angry, Fearful, Disgust, and Sad emotion as negative. The merged data (CREMA-D and RAVDESS) was divided into 70% training, 20% validation, and 10% testing, for proper learning and testing.

Some preprocessing is involved in the data to get prepared. Raw waveform were first normalized to the same RMS and then down sampled if necessary to 16 kHz. Background noise was filtered by low-pass filters, which excluded the undesired noise signal, thus improved the signal-to-noise ratio. The audio signals were then converted into a feature representation of 50 Mel-frequency cepstral coefficients (MFCCs) per frame, which have been shown to be efficient in speech and emotion recognition tasks [7].

B. Model Architecture and Training

For this project a hybrid CNN and BiLSTM architecture has been chosen to efficiently learn spatial and temporal features from the MFCC input. The CNN layers are used as local pattern extractors from the MFCCS and the BiLSTM layers capture long term temporal dependencies in user's speech, which are essential to properly recognize an user emotional tone [8].

The hybrid CNN-BiLSTM model consists of two convolutional layers (64 and 128 filters respectively, each with a kernel size of 3x3), followed by batch normalization, max pooling, and dropout layers (30%). These layers serve to extract spatial patterns from the MFCC input of shape (139, 50, 1). The output is reshaped into (34, 1536) and passed to a bidirectional LSTM layer (128 units) that captures temporal dependencies in both forward and backward directions. The final layers include a fully connected dense layer with 128 neurons and ReLU activation, followed by a softmax output layer that classifies the input into one of three emotion classes.

TABLE I. MODEL ARCHITECTURE OF CNN-BILSTM

Layer No.	Layer Type	Parameters / Configuration	Output Shape (Approximate)
1	Input Layer	Input shape = (139, 50, 1)	(139, 50, 1)
2	Conv2D	64 filters, kernel size = (3,3), activation = ReLU, padding = 'same'	(139, 50, 64)
3	BatchNormalization	—	(139, 50, 64)
4	MaxPooling2D	Pool size = (2,2)	(69, 25, 64)
5	Dropout	Rate = 0.3	(69, 25, 64)
6	Conv2D	128 filters, kernel size = (3,3), activation = ReLU, padding = 'same'	(69, 25, 128)
7	BatchNormalization	—	(69, 25, 128)
8	MaxPooling2D	Pool size = (2,2)	(34, 12, 128)
9	Dropout	Rate = 0.3	(34, 12, 128)
10	Reshape	Reshape to (34, 1536)	(34, 1536)
11	Bidirectional LSTM	128 units, return_sequences=False	(256)
12	Dropout	Rate = 0.5	(256)
13	Dense	128 units, activation = ReLU	(128)
14	Dropout	Rate = 0.3	(128)
15	Output (Dense)	3 units (softmax), classes: positive, neutral, negative	(3)

The loss was formulated as the categorical cross-entropy and the model was optimized using Adam. The dataset was split into 70% training, 10% validation, and 20% testing sets. Evaluation was performed on the testing subset, separate from training, and performance metrics (accuracy, precision, recall, F1 score) were calculated. No cross-validation was used due to time constraints, but stratified splitting was maintained to ensure balanced emotion distribution. The model had high accuracy and low loss during training and then was converted to a optimized lighter version of the model using TensorFlow Lite for mobile use.

C. Android Application Architetcure

Flutter is used for the app's cross-platform nature. But practical placement issues like incompatibility with on-device real-time TensorFlow Lite inference and audio feature extraction caused them to settle back to native apps through Android Studio and Java. This change made it possible to integrate TensorFlow Lite and the TarsosDSP library to generate feed of MFCCs in real-time.

The app was developed in Android Studio and it consisted in three modules, one to record voice that captures the users thoughts, feelings, emotions, etc., other to predict the user emotion type using the deep learning prediction model, and the last module to recommend songs to the user based on predicted emotions. Turning to the system architecture, we have the microphone interface, a real-time MFCC extractor, an inference module, and a media player which adjusts its playlist according to the emotions it detects.

D. UI and UX Design

Ease of use and simple interface makes the mobile application overwhelmingly popular. The primary screen has a waveform which visualizes real - time voice input, an interactive button to start the recording and analysis, and emotion - specific background themes which are adjustable according to the detected emotion, for example, warm colour for positive, grayscale for neutral, and dark/cool colour for negative. These UI modifications maintained the mood of the recommended music, aiding flow of experience.

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E. Speech Processing and Emotion Recognition Pipeline

After the user starts recording, the TarsosDSP library computes MFCCs on the raw audio on-the-fly. They are fed into the C NN -BiLSTM model, which classifies emotion into positive, neutral and negative sentiment. The mobile based inference time was finely tuned to be less than 1.5 seconds with a good compromise to the real-time experience.

F. Music Recommendation and Playback

After a successful emotion classification, the system plays music from a local playlist with predefined preferences of music ordered by similar emotion states. During creation, songs are labelled by the emotion they evoke, using variables such as speed (BPM), key and attributes of timbre. For example, positive emotions provoke an energetic, fast-paced melody in major keys, whereas negative emotions elicit a more relaxed melody in minor keys [9], [10]. The music player automatically plays the matched song for easy, seamless speech analysis

G. Evaluation and User Testing

The system was evaluated through structured user testing involving 20 participants aged 20–40 from diverse linguistic backgrounds. Participants were instructed to record speech under three conditions: quiet room, driving, and noisy background. Each user tested the system at least three times per scenario and shared their feedback using a Google Form survey. Both emotion recognition accuracy, quality of playback, and overall satisfaction were evaluated.

Participants were informed of data privacy and consent. The app does not store or transmit audio recordings; all processing occurs on-device. Users agreed to a testing consent form detailing the process and privacy.

For driving tests, users were instructed to operate the app only when the vehicle was stationary or operated by a co-driver. A safety disclaimer was displayed at app startup.

Technical performance was evaluated using confusion matrices and recognition rates; usability was measured by means of Likert scale responses [11].

III. System Architecture

The architecture of the proposed system is intended to function end-to-end as a real-time pipeline for capturing the user's speech, processing it, classifying it in terms of the expression of the user's own emotions, and then providing a recommendation of music which is considered to be consistent with the emotional state found to be expressed in the user's speech. In this section, we provide an overview of the fundamental design components of the mobile app: the voice processing engine, GUI, inference engine, and song recommendation logic.

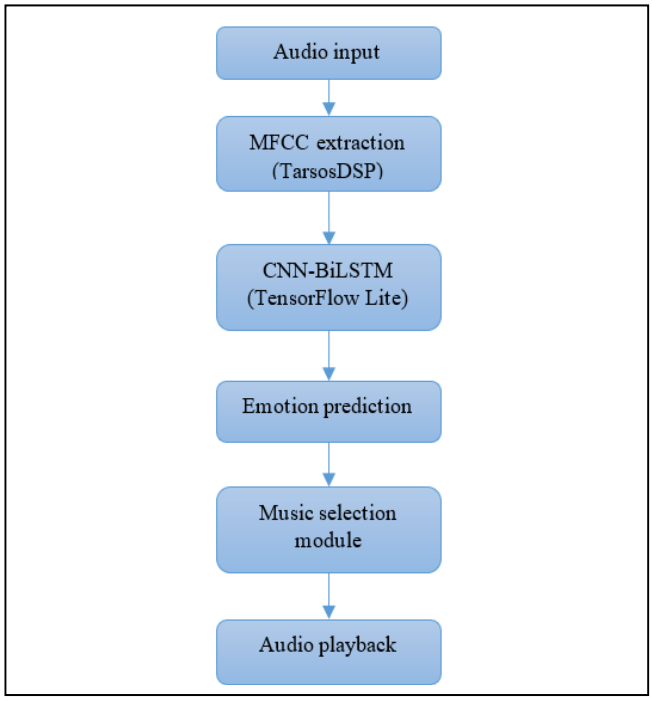

Fig. 1. End-to-End Pipeline of the Emotion-Based Music Recommendation System.

A. Real-Time Voice Capture and Feature Extraction

The Android app uses the microphone of the user's device to record speech inputs. In order to extract meaningful emotional characteristics in the audio signal in real-time, the app makes use of the TarsosDSP library, a real-time digital signal processing toolbox specially designed for mobile devices. This library makes calculation of Mel-Frequency Cepstral Coefficients (MFCCs) on-the-fly while a speech signal is being recorded.

The extracted MFCC features are used as input to the deep embedded emotion classification model. They are also applicable for describing nuances of emotions in voice as they are of human speech and they rely on their FFT frequency based dimensionality [12].

B. On-Device Emotion Inference Using Deep Learning

The trained CNN-BiLSTM model was converted to TensorFlow Lite format to work in real-time. This optimization enables the model to run inference natively on Android devices without introducing significant delays and having satisfactory user experience.

The inference module takes MFCC vectors as input, feeds them through the model, and produces a prediction among positive, neutral, and negative. The total execution time from feature extraction to the final diagnosis in all the pipelines is less than 1.5 seconds, which makes the system response in real time enough for usability.

C. User Interface and Emotion Feedback

The UI of the application is simple, responsive and comparable with feed of emotional feedback. The core UI elements include:

i. A microphone button on the centre for voice recording.

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ii. Realtime audio input is visualized by a waveform visualizer.

iii. A colour-changing emotion-sensitive background colour

This proposed design not only provides strong UX by bolstering what is detected, but also imparts a strong personalized feeling. Furthermore, the interface is designed for single handed use and tested on different screen sizes for accessibility.

D. Emotion-Driven Music Recommendation Logic

After the mood of the user is determined, the system will play back an emotion matching music track from the locally stored playlist. Songs are pre-segmented according to their emotional tone on musical features such as:

i. Tempo (Beats Per Minute – BPM)
ii. Key signature (Major for positive, Minor for negative)
iii. Timbre and rhythm structure

This approach allows the recommended music to correspond to the user's present mood. For example, if the system recognizes a positive emotion, it selects an high-arousal, major-key song; for neutral, it chooses ambient or low-energy music; and negative, it selects calm, low-tempo songs that may induce positive affect or have valence congruency [9], [10], [13].

IV. Results and Discussion

This section demonstrates the technical evaluation and user validation of a real-time SER system. Results are presented quantitatively evaluating the model on various datasets, the efficiency of mobile deployment and the recommendation effectiveness of the emotion-based music recommendation system. The paper also discusses feedback from real world user testing to evaluate application in terms of usability, accuracy and emotional responsiveness.

A. Dataset Analysis and Model Training Results

The CNN-BiLSTM model was trained using the combined TESS, CREMA-D, and RAVDESS datasets. Each dataset underwent preprocessing to ensure compatibility, such as converting stereo files to mono and extracting consistent MFCC features. Separate experiments were conducted to evaluate model accuracy across individual and combined datasets.

The confusion matrix for the TESS dataset shows a high classification rate for the neutral emotion, while positive and negative emotions exhibited occasional misclassifications due to overlapping vocal cues. When combining CREMA-D and RAVDESS, classification performance improved, especially for negative emotions, due to the diversity in vocal expressions and speaker characteristics.

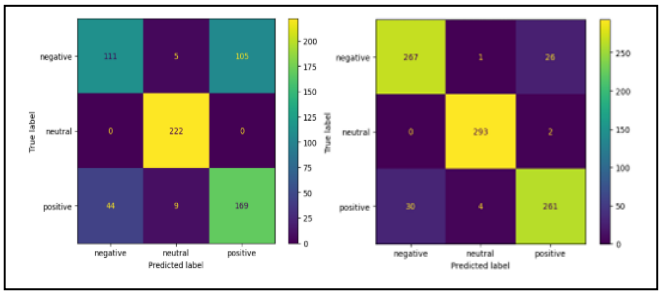

Fig. 2. Confusion Matric of TESS (left) and RAVDESS + CREMA-D (right).

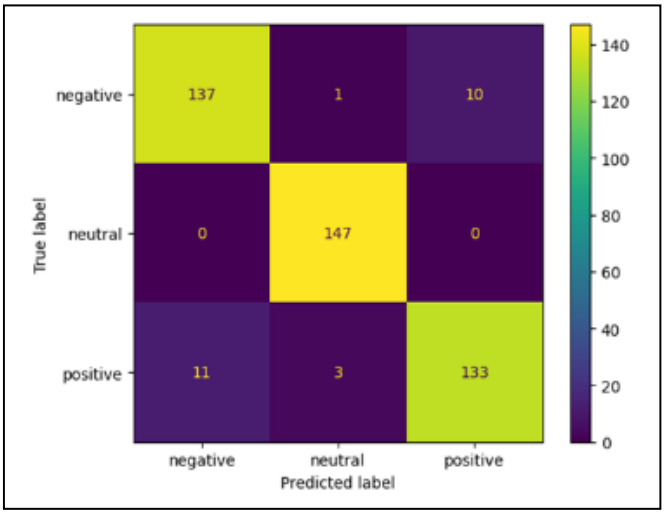

Fig. 3. Confusion Matric of Final Trained Test Set.

Classification Report:

	precision	recall	f1-score	support
negative	0.93	0.93	0.93	148
neutral	0.97	1.00	0.99	147
positive	0.93	0.90	0.92	147
accuracy			0.94	442
macro avg	0.94	0.94	0.94	442
weighted avg	0.94	0.94	0.94	442

Fig. 4. Classification Report of Final Trained Test Set.

B. Deployment Constraints and Model Optimization

Initially, deployment via Flutter was considered due to its cross-platform development advantages. However, real-time MFCC processing and TensorFlow Lite model inference proved difficult in Flutter due to limited support for native audio processing and machine learning inference libraries. As a result, the development was transitioned to native Android Studio using Java, which allowed for seamless integration of TarsosDSP for feature extraction and TensorFlow Lite for on-device inference.

Model size and complexity were optimized to balance performance and resource usage. The final TensorFlow Lite model was under 5 MB, ensuring efficient loading times and compatibility with mid-range Android devices.

C. Application Interface and Real-Time Behavior

The application was tested in different using scenarios and environments. Waveform drew the real-time voice input and app UI was updated on-the-fly with mood-specific themes according to the prediction results.

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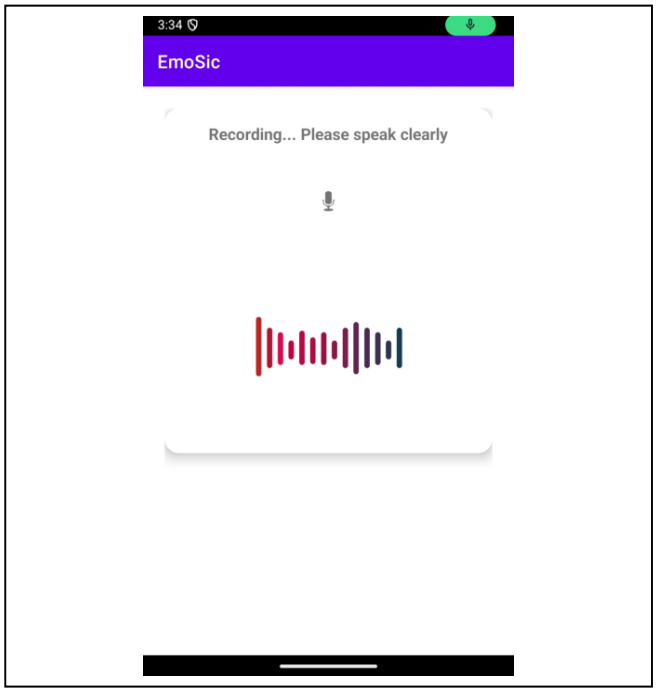

Fig. 5. Waveform for real time input.

Final emotion prediction results were flashed for a second on the screen before the display changed to the media player where a congruent song streamed from the predetermined playlist. Everything from speech input to music playback in the end-to-end workflow itself were optimized for a continuous dynamic dialogue.

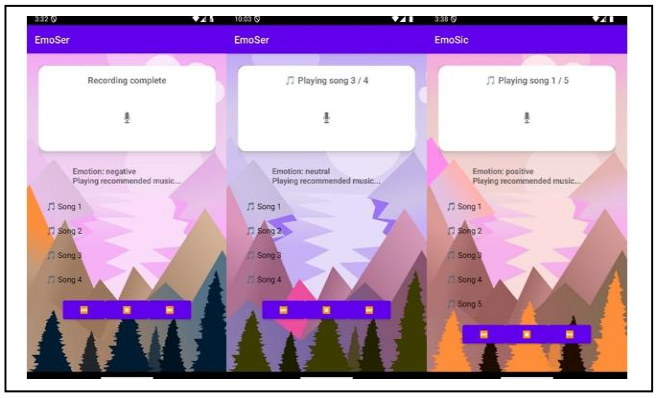

Fig. 6. Background of different emotion detected.

D. Evaluation: User Testing via Google Form

To validate system effectiveness, **user testing** was conducted among participants in three settings: a quiet room, while driving, and in noisy environments. Participants were asked to record short voice inputs, observe the predicted emotion, and evaluate the corresponding music playback.
Key results from user feedback included:

- Emotion Recognition Accuracy: 75.1% average success rate across all test cases.
- Playback Relevance: 87.5% of users found the recommended music aligned with their emotional expectations.
- System Usability: 100% of users rated the app as "Very Easy to Use."
- Overall Rating: Average app rating of **4.5/5** based on user survey

E. Music Playback Strategy and Observed Limitations

The current music recommendation logic successfully delivers emotion-appropriate content based on pre-tagged local audio files. However, limitations were observed:

- The system is limited to three emotion categories: positive, neutral, and negative.
- The application currently does not support all language emotion detection, which limits its effectiveness for other speaker.
- The data were restricted to acted emotional speech, and the user testing involved only 20 test participants, so generalization of the results might be compromised.

Additionally, music files are stored locally, restricting song variety. There is potential to expand functionality by integrating external APIs for example Spotify and YouTube, and building a dynamic, cloud-based playlist system.

V. Conclusion and Future Work

This paper presents the design, implementation, and evaluation of a real-time Speech Emotion Recognition (SER) system integrated into a native Android mobile application for emotion-based music recommendation. The primary objective was to bridge the gap between user emotional state and music personalization by leveraging deep learning techniques capable of understanding vocal expressions in real-time.

The system utilizes a hybrid CNN-BiLSTM model trained on RAVDESS, and CREMA-D datasets to classify speech into three basic emotional states: positive, neutral, and negative. Preprocessing steps such as silence trimming, noise filtering, and MFCC feature extraction were performed to ensure high-quality input. The model demonstrated strong performance with an average recognition accuracy of 93% on test data and 90% recognition success in real-world user testing.

To support mobile deployment, the model was converted into TensorFlow Lite, and the app was redeveloped in native Android using Java. This transition addressed compatibility issues experienced in Flutter and enabled smooth integration of TarsosDSP for real-time MFCC extraction. The final application, named EmoSer, provides an intuitive user interface and dynamic music playback based on the detected emotion.

User feedback collected via Google Form surveys revealed a high level of satisfaction with both emotion recognition and music playback accuracy. Users particularly appreciated the simplicity of the interface and the seamless experience of transitioning from voice analysis to music recommendation. However, the system also revealed several limitations, including:

- Restriction to only three broad emotion categories.
- Local music dependency with no streaming integration.
- Sensitivity to background noise in uncontrolled environments.

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To address these limitations and further enhance the system, the following future work is proposed:

1. Multilingual Support: Extend the emotion recognition capabilities to multiple languages and accents, increasing the system's accessibility to a wider audience.
2. Noise Robustness Enhancement: Implement more advanced noise cancellation and adaptive filtering methods to improve emotion detection accuracy in noisy or unpredictable environments.

While the current system detects three broad emotional states (positive, neutral, negative), expanding the classification to include emotions such as anger, sadness, and excitement would significantly improve personalization and generalizability.

In conclusion, this study has demonstrated that speech emotion recognition can be effectively used to personalize music experiences on mobile devices in real-time. With further development, this technology has the potential to support emotionally intelligent applications across wellness, entertainment, and assistive domains.

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